Problem Set 1
ECO3121 - Fall 2024

Due 3 PM, October, 11, 2024
Please remember to submit your **Stata code** for **Question 2** (including **".do" file** that records commands and **".log" file** that records corresponding outputs) as it will be graded.

Question 1

Suppose we are given a sequence of data $\{y_i, x_i\}_{i=1}^n$. We are trying to estimate two models.

1. Consider the following model: $y_i = \beta_0 + \mu_i$. Derive the OLS estimator of β_0. (2 points)
2. Consider the following model: $y_i = \beta_1 x_i + \mu_i$. Derive the OLS estimator of β_1. (2 points)

Question 2

Download from the blackboard site and load into Stata the Indonesia education and earning data `inpresdata.dta`. The main data we use is the 1995 intercensal survey of Indonesia (SUPAS), which is a nationally representative large cross section of men born between 1950 and 1972 from the 1995 intercensal survey of Indonesia. It is the main dataset used in the influential paper "Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment" by Esther Duflo.

Since this will be a univariate model, we'll be estimating regressions of the form:

$$wage_i = \alpha + \beta_1 education_i + \mu_i$$

First, let's analyze the effect of **years of education** on **monthly wages**.

1. What sign do you predict β_1 will have? Why? (1 point)
2. Run the regression and report the estimated coefficients, their standard errors, and R^2. Does the estimated value of β_1 agree with your prediction? (2 points)
3. Interpret the value of the estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ from the model. (2 points)
4. Compute the fitted value $\widehat{wage}_i$ and the residual $\hat{\mu}_i$ for each observation, and verify that the residuals (approximately) sum to 0. (1 point)
5. According to the estimated relation, what is the predicted $\widehat{wage}_i$ for an individual with an education level of 10 years? (1 point)

6. How much of the variation in wage for these individuals is explained by their education years? Can you provide some reasons for why education might be lower among rural residents? (2 points)

7. A professor asks you whether they should estimate a linear relationship between wage and education. She asks you to explore alternative functional forms for this relationship. Using the provided data, you estimate the following equations:
   * A log-linear relationship: $log(wage)_i = \beta_0 + \beta_1\, education_i + \mu_i$
   * A linear-log relationship: $wage_i = \beta_0 + \beta_1\, log(education)_i + \mu_i$
   * A log-log relationship: $log(wage)_i = \beta_0 + \beta_1\, log(education)_i + \mu_i$

   ```
   We take natural log here. In Stata, the coding is "log(variable)"
   ```

   Run the regressions recommended by the professor. Interpret the parameters on education years (or log(education years)) in each of these equations. (3 points)

8. Describe the SLR.4 assumption under this case. Do you think that assumption holds? (1 point)

Question 3

A large-landholding farmer outsmarts others by running an experiment to measure the effect of fertilizer application on agricultural yields. She **randomly assigns** her 400 land plots into two groups. In the first group, she inputs 20 kg/mu of fertilizers, while the other group receives 40 kg/mu of fertilizers. She also keeps other agricultural inputs (water, pesticides, labors) equally between these two groups. Let Y_i denote the yields for i^{th} plot, let X_i denote the amount of fertilizer application per unit of land, ($X_i = 20$ or 40), and consider the regression model $Y_i = \beta_0 + \beta_1 X_i + \mu_i$

1. Explain what the error term μ_i represents. Propose a potential reason that why different plots might have different values of μ_i? (2 points)

2. Is $E[\mu_i|X_i] = 0$ in this case? Explain why or why not? (1 point)

3. The estimated regression is $Y_i = 300 + 0.7X_i + \mu_i$.

   (a) Is $\hat{\beta}_1 = 0.7$ an unbiased estimator? (1 point)

   (b) After you got this estimation, the farmer's assistant tells you that he is concerned that the program may not actually have randomly allocated fertilizer quantity across the 400 plots, and that some cheating may have gone on (he heard that the richest land (land with highest soil quality) were more likely to be put into the more fertilizer group). What concern would this give rise to in your estimation? (1 point)

   (c) What additional variable (data) would you like to collect to verify whether the assistant's concern is true, and what regression specification would you use these data for to investigate whether his concern is true? (3 points)